

## THERMACART · DESIGN DECISION RECORD

# ThermaCart recommendations accepted

Design Molecule Lab · designmolecule.com

## DOCUMENT CONTROL

|             |                                                               |         |                |
|-------------|---------------------------------------------------------------|---------|----------------|
| Document ID | TCT-DDR-002                                                   | Version | 0.1            |
| Status      | Draft                                                         | Date    | 2026-09-25     |
| Author      | Amish Chadha                                                  | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept                                        |         |                |
| Repository  | github.com/BoujeeEnjinia1701/thermacart                       |         |                |
| Source      | docs/decisions/0002-recommendations-accepted.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                         |
|---------|------------|--------------|----------------------------------------------------------------------------------------------------------------|
| 0.1     | 2026-09-25 | Amish Chadha | Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo, and the items still open |

## 0002: Recommendations accepted

---

- **Date:** 2026-09-25
- **Status:** accepted. The front matter status stays Draft because the documentation standard allows only Draft, In review, Released or Superseded.

### ■ Context

The TRL 3 session recorded items D1 to D8 and T1 to T4 in TCT-DDR-001 as "adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review", and items O1 to O9 as "Proposed, awaiting Amish" ( docs/REVIEW.md , session 2026-09-25, TRL 3). On 2026-09-25 Amish wrote in chat: "i accept all your recommendations, go with them across all repos." Every item that carries a recommendation is therefore decided, and where a recommendation named one of several options, that option is the decision. Items without a recommendation stay open.

### ■ Decision

*Table 1. Items decided by Amish, 2026-09-25: go with recommendation.*

| #        | Item                  | Decision                                                                                                                           | What changed in the repo                                                                                                                                                                                                                                                                                                                     |
|----------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1       | Envelope              | Stock 6 x 2 x 1/8 in tube in a GN 1/3 envelope; WHO PQS 0.6 L size later; extrusion only at volume                                 | Status wording only; the design already followed it                                                                                                                                                                                                                                                                                          |
| D2       | Grades                | C5, C25 and H70 paraffins, W0 produce-only option, C5 first                                                                        | Status wording only                                                                                                                                                                                                                                                                                                                          |
| D3       | Pitch                 | "Cold chain, comfort and hot holding"                                                                                              | Already applied to <code>project.yaml</code> and <code>README.md</code> ; no further change                                                                                                                                                                                                                                                  |
| D4       | Keying and indicator  | Grade nose key, published key table, sight window                                                                                  | Status wording only                                                                                                                                                                                                                                                                                                                          |
| D5       | 2 mm wall             | Evaluated and not adopted                                                                                                          | Status wording only                                                                                                                                                                                                                                                                                                                          |
| D6       | Budget                | Keep <code>budget_usd</code> at \$80 for one C5 cartridge and one frame                                                            | <code>budget_usd</code> unchanged at \$80                                                                                                                                                                                                                                                                                                    |
| D7       | First host type       | A produce cooler user through a local partner                                                                                      | Status wording only; no partner named (O1)                                                                                                                                                                                                                                                                                                   |
| D8       | Fill and shell        | Fill as liquid at the grade limit, seal for life, aluminium shell, no HDPE                                                         | Status wording only                                                                                                                                                                                                                                                                                                                          |
| T1 to T4 | TRL 3 refinements     | Folding bail on a thermal break, cap stack from plate, stock fin bar, 85 °C oven limit                                             | Status wording only                                                                                                                                                                                                                                                                                                                          |
| O4       | R5, PCM fraction      | Option (a): relax R5 to 38 % for the stock-tube build, with (c) 50 % kept as the target for a later extrusion                      | R5 in TCT-REQ-001 v0.4 restated; status from not met (38.3 to 39.9 % against 50 %) to met, thin (38.1 to 39.7 % against 38 %)                                                                                                                                                                                                                |
| O5       | R10, cold hold        | Option (a): dark finish plus user guidance to lay produce on the cartridges; R10 kept at 8 h until a test shows the contact effect | Guidance added to TCT-PRC-001 v0.4 (How it works); R10 target unchanged; hold 1.9 h (bare, v0.1 basis) to 2.4 h (dark, now the design case), still not met. The measurement is TRL 4 work: decided but on hold                                                                                                                               |
| O6       | R16 and budget        | Option (b): get supplier quotes first; do not raise the budget until quotes are in                                                 | <code>budget_usd</code> stays \$80. Getting quotes is a purchasing step: decided but on hold with TRL 4. R16 stays not met                                                                                                                                                                                                                   |
| O7       | R14, H70 from an oven | Adopt a handling rule on the label (oven gloves or a 12 min wait) and prefer pad charging                                          | R14 restated with the rule; status from at risk to met with the handling rule; H70 label text, precis, safety text and drawing note updated                                                                                                                                                                                                  |
| O8       | Surface finish        | Dark finish                                                                                                                        | Row 1 of <code>bom/bom.csv</code> adds etch primer and matte black high-temperature paint (\$24 to \$29); <code>finish</code> parameter in <code>cad/src/model.py</code> ; TCT-DWG-001 Rev P1 to P2 with a finish note; media re-rendered with a black shell; TCT-CAL-001 v0.2 uses emissivity 0.9 as the design case and adds 15 g of paint |
| O9       | R1, C25 band          |                                                                                                                                    | R1 restated; status from at risk to met                                                                                                                                                                                                                                                                                                      |

| # | Item | Decision                          | What changed in the repo |
|---|------|-----------------------------------|--------------------------|
|   |      | Widen the C25 band to 22 to 27 °C |                          |

## ■ Effect on the numbers

Table 2. Numbers before and after (TCT-CAL-001 v0.1 to v0.2).

| Quantity                                                 | Before                                        | After                                                                         |
|----------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------|
| Cartridge parts cost                                     | \$71.00                                       | \$76.00                                                                       |
| One C5 cartridge and one frame, against the \$80 budget  | \$87.00 (\$7 over)                            | \$92.00 (\$12 over)                                                           |
| Three-grade set plus one frame                           | about \$230                                   | about \$245                                                                   |
| Empty cartridge                                          | 1.78 kg                                       | 1.80 kg                                                                       |
| Filled mass                                              | 2.89 to 2.96 kg                               | 2.90 to 2.98 kg (R4 still met, thin)                                          |
| PCM fraction                                             | 38.3 to 39.9 %                                | 38.1 to 39.7 %                                                                |
| R10 hold, two C5 at 32 °C (design case)                  | 1.9 h bare                                    | 2.4 h dark                                                                    |
| R11 hold, one H70 (design case)                          | 7.2 h bare                                    | 13.6 h dark                                                                   |
| R12 C5 freezer, C25 refrigerator, H70 oven (design case) | 5.6, 6.6 and 6.1 h                            | 4.2, 4.6 and 5.3 h                                                            |
| Requirement counts                                       | 3 not met, 3 at risk, 3 not verifiable, 9 met | 2 not met (R10, R16), 1 at risk (R12), 3 not verifiable (R6, R7, R17), 12 met |

## ■ Items still open

Table 3. Items with no recommendation, still "Proposed, awaiting Amish".

| #  | Item                                                                                                                                                                         |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| O1 | The named co-design partner and region for the first produce cooler host                                                                                                     |
| O2 | Whether ColdPod or ZeerBox should declare ThermaCart as a dependency                                                                                                         |
| O3 | C5 charging: freezer with a conditioning step, or a refrigerator (TCT-CAL-001 shows 31 to 51 h in a refrigerator, which favours the freezer, but no recommendation was made) |

## ■ Consequences

- Controlled documents bumped: TCT-PRB-001 v0.4, TCT-PRC-001 v0.4, TCT-REQ-001 v0.4, TCT-CAL-001 v0.2, TCT-DDR-001 v0.2. Drawing TCT-DWG-001 is at Rev P2.
- `project.yaml`: `budget_usd` stays at \$80, `trl`: 3 and `trl_target`: 3.
- No cross-repo action follows from these decisions; O2 has no recommendation and no other repo was changed.

- The R10 contact measurement and supplier quotes are TRL 4 work, on hold by Amish's instruction. Nothing in this record authorizes building, testing or purchasing.